

Results Interpretation

Research questions: How has malaria incidence changed across countries from 2000 to 2024, and which countries experienced the largest reductions and increases?

Objectives: 1. To examine how malaria incidence has changed across countries from 2000 to 2024. 2. To determine which countries experienced the largest reductions or increases.

Results

This study included 108 countries but only 103 countries had complete data for the years 2000 and 2024. The results showed overall, reduction in malaria between 2000 and 2024 across 103 countries (median = -17.78, IQR = 109.76). A Wilcoxon Signed Rank Test was used to compare the difference in malaria incidence in 2000 and 2024. A statistically significant difference was observed in the results p-value < 0.001. The median of malaria incidence in 2024 (median = 7.1991, IQR = 179.67) was lower than the median of malaria incidence in 2000 (median = 31.7598, IQR = 325.95) (Table 1).

However, the change was not uniform across countries. The largest reduction in malaria incidence was observed in Solomon Islands, Guinea-Bissau and Burkina Faso with changes of -359.54, -286.34 and -243.68 cases per 1000 population, respectively. In contrast, Madagascar, Eritrea and Djibouti experienced the largest increases, with changes of 200.59, 83.43, and 42.02 cases per 1,000 population, respectively (Figure 1) (Table 2).

Table 1: Malaria incidence from 2000 to 2024.

	Median	IQR	p-value	z-score
Malaria incidence in 2000	31.7598	325.95	-	-
Malaria incidence in 2024	7.1991	179.67	-	-
Change in malaria incidence	-17.7818	109.76	< 0.001	-7.653

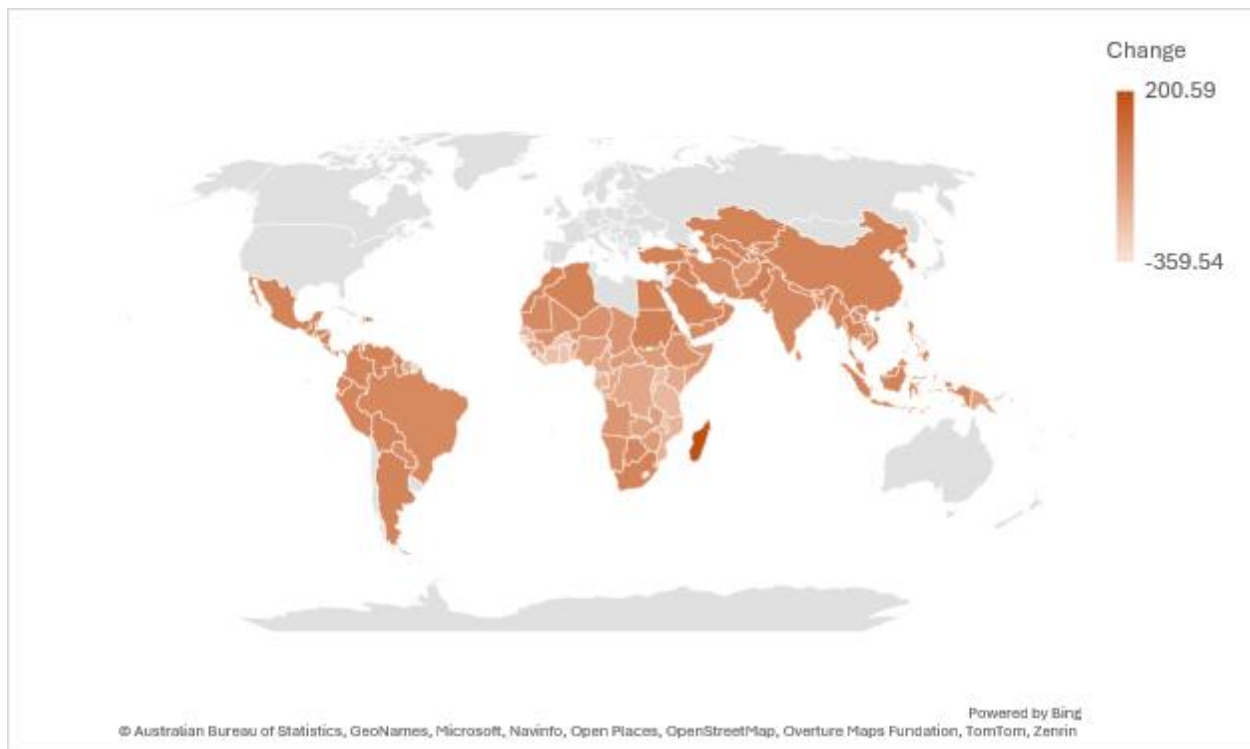


Figure 1: Change in malaria incidence rate across 103 countries between 2000 and 2024.

Table 2: Malaria incidence in 2000 and 2024 across 108 countries, organized by the difference in incidence from the highest increase to the highest decrease.

	Country	Incidence in 2000 - rate per 1000	Incidence in 2024 - rate per 1000	Change in malaria incidence - rate per 1000
1	Madagascar	54.854	255.4457	200.5917
2	Eritrea	18.9988	102.4244	83.4255
3	Djibouti	3.0727	45.0946	42.0219
4	Sudan	89.7208	98.2517	8.531
5	Pakistan	6.0848	12.7716	6.6868
6	Venezuela (Bolivarian Republic of)	2.8962	7.7881	4.8919
7	Panama	0.5409	3.5036	2.9627
8	Egypt	0	0	0
9	Morocco	0.0001	0	-0.0001
10	Syrian Arab Republic	0.0004	0	-0.0004
11	Kyrgyzstan	0.0018	0	-0.0018
12	Oman	0.0026	0	-0.0026
13	China	0.0151	0	-0.0151
14	Algeria	0.0187	0	-0.0187
15	Armenia	0.0451	0	-0.0451
16	Turkmenistan	0.0806	0	-0.0806

17	Dominican Republic	0.3223	0.1911	-0.1312
18	Iraq	0.5858	0	-0.5858
19	El Salvador	0.6241	0	-0.6241
20	Haiti	9.7137	8.6958	-1.0179
21	Cabo Verde	1.2218	0.1392	-1.0825
22	Republic of Korea	1.2778	0.1768	-1.101
23	Costa Rica	1.3634	0.1337	-1.2297
24	Philippines	1.7305	0.3531	-1.3774
25	Eswatini	2.7024	0.5546	-2.1478
26	Argentina	2.3647	0	-2.3647
27	Indonesia	5.2839	2.2267	-3.0572
28	Mexico	3.5349	0.0959	-3.439
29	Viet Nam	3.5422	0.0032	-3.539
30	Comoros	66.3708	62.787	-3.5838
31	South Africa	3.8597	0.1148	-3.7448
32	Saudi Arabia	3.8092	0	-3.8092
33	Uzbekistan	5.0833	0	-5.0833
34	Guyana	43.9678	38.7409	-5.2269
35	Georgia	5.6604	0	-5.6604
36	Nicaragua	6.7768	1.0437	-5.7332
37	Bangladesh	6.6167	0.8605	-5.7562
38	Thailand	6.5706	0.5793	-5.9914
39	Nepal	6.7314	0.0043	-6.7271
40	Guatemala	7.2107	0.2105	-7.0002
41	Bolivia (Plurinational State of)	11.6911	3.8284	-7.8626
42	Honduras	8.6433	0.2337	-8.4096
43	Belize	8.9437	0	-8.9437
44	Tajikistan	9.082	0	-9.082
45	Colombia	24.3491	15.1915	-9.1576
46	Democratic People's Republic of Korea	9.7907	0.5052	-9.2856
47	Peru	13.1889	3.245	-9.9439
48	Bhutan	13.4374	0	-13.4374
49	Malaysia	13.8291	0	-13.8291
50	Ecuador	14.536	0.037	-14.499
51	Botswana	17.2255	0.2004	-17.0251
52	Brazil	21.5356	3.7538	-17.7818
53	Myanmar	34.4025	14.7795	-19.623
54	India	21.8034	1.4806	-20.3228
55	Mauritania	96.394	69.5536	-26.8404
56	Iran (Islamic Republic of)	29.0739	2.1756	-26.8983
57	Yemen	83.2982	52.1692	-31.129
58	Lao People's Democratic Republic	31.7598	0.1331	-31.6267

59	Paraguay	37.3194	0	-37.3194
60	Namibia	53.57	7.1991	-46.3709
61	Sri Lanka	47.3338	0	-47.3338
62	Mali	394.977	346.2196	-48.7574
63	Ethiopia	189.1844	138.495	-50.6893
64	Niger	359.2388	305.1436	-54.0952
65	Benin	416.0037	354.2531	-61.7506
66	Chad	272.1289	206.1381	-65.9908
67	Afghanistan	84.4758	17.7627	-66.7131
68	Angola	326.5394	258.8999	-67.6395
69	Cambodia	75.4731	0.0257	-75.4475
70	Somalia	133.1488	53.2775	-79.8713
71	South Sudan	344.5879	254.0781	-90.5097
72	Zimbabwe	102.5884	11.3627	-91.2257
73	Central African Republic	438.1199	343.8437	-94.2762
74	Rwanda	180.4321	77.1384	-103.294
75	Equatorial Guinea	334.3012	227.7302	-106.571
76	Nigeria	401.7884	294.2518	-107.537
77	Zambia	360.3289	252.4103	-107.919
78	Vanuatu	124.3659	12.2461	-112.12
79	Papua New Guinea	264.0839	151.7626	-112.321
80	Burundi	427.4814	314.8994	-112.582
81	Congo	349.6006	221.0209	-128.58
82	Guinea	415.775	286.0837	-129.691
83	Cameroon	394.8476	260.4595	-134.388
84	Democratic Republic of the Congo	458.2442	321.893	-136.351
85	Sierra Leone	424.5773	282.6657	-141.912
86	Kenya	220.2575	74.1683	-146.089
87	Malawi	452.5498	294.528	-158.022
88	Timor-Leste	158.379	0	-158.379
89	Suriname	161.3778	0	-161.378
90	Gabon	364.6591	187.002	-177.657
91	Mozambique	473.668	295.0967	-178.571
92	Sao Tome and Principe	222.0209	30.0888	-191.932
93	Senegal	230.3672	36.7807	-193.587
94	Togo	445.1167	249.2522	-195.865
95	United Republic of Tanzania	334.6066	136.7251	-197.882
96	Uganda	482.7711	264.2318	-218.539
97	Côte d'Ivoire	490.5169	267.9491	-222.568
98	Liberia	397.0758	172.3411	-224.735
99	Gambia	309.7216	75.4442	-234.277
100	Ghana	430.3574	195.7697	-234.588
101	Burkina Faso	597.1736	353.4889	-243.685

102	Guinea-Bissau	396.4632	110.1278	-286.335
103	Solomon Islands	583.818	224.2795	-359.539
104	Azerbaijan*	8.1162	-	-
105	Kazakhstan*	-	0	-
106	Mauritius*	-	-	-
107	Türkiye*	2.6882	-	-
108	United Arab Emirates*	-	0	-

* Countries with missing incidence rate in either 2000, 2024 or both.